TSHR and IGF1R tissue results

Fixed panel of five GTEx v8 tissues and five disease datasets

A Instrument availability

Four of ten gene–tissue combinations yielded one qualifying instrument each. The remaining six combinations were unestimable; they were retained in the planned testing denominators.

Gene	Tissue	N	IVs	F	Minimum eQTL P	Availability
TSHR	Thyroid	574	0	—	9.60×10^{-8}	No eQTL at P < 5×10^{-8}
IGF1R	Thyroid	574	1	61.1	3.14×10^{-14}	One SNP retained
TSHR	Cultured fibroblasts	483	0	—	—	Gene absent from tested-gene file
IGF1R	Cultured fibroblasts	483	1	33.8	1.23×10^{-8}	One SNP retained
TSHR	Subcutaneous adipose	581	1	38.5	1.12×10^{-9}	One SNP retained
IGF1R	Subcutaneous adipose	581	0	—	3.56×10^{-7}	No eQTL at P < 5×10^{-8}
TSHR	Visceral adipose	469	1	32.1	2.77×10^{-8}	One SNP retained
IGF1R	Visceral adipose	469	0	—	7.76×10^{-6}	No eQTL at P < 5×10^{-8}
TSHR	Whole blood	670	0	—	2.46×10^{-4}	No eQTL at P < 5×10^{-8}
IGF1R	Whole blood	670	0	—	5.42×10^{-5}	No eQTL at P < 5×10^{-8}

N is the GTEx tissue eQTL sample size. IVs are retained instrumental genetic variants. Eligibility required cis ±1 Mb, eQTL P < 5×10^{-8}, F > 10, MAF ≥ 0.01, and EUR-reference LD clumping at $r^2 < 0.001$ within 10 Mb. No threshold was relaxed.

The thyroid TSHR minimum eQTL P was 9.60474×10^{-8} and did not qualify. The cultured-fibroblast TSHR gene was absent from the tested-gene file. Neither situation is a negative MR result. Cultured fibroblasts and adipose tissues are not orbital tissue. GTEx whole blood (N = 670) is distinct from the original eQTLGen exposure (up to 31,684 participants); lack of a GTEx instrument does not invalidate the original blood analysis.

Planned tests: ten primary comparisons against FinnGen GD, and forty secondary comparisons against the other four datasets. Twenty estimates were available; all thirty unestimable comparisons remain in Supplementary Dataset, Tissue_extension.

B All estimable disease associations

All estimates use one SNP and the Wald ratio. FinnGen GD is the primary comparison within this post hoc tissue extension, not a new primary endpoint of the original study.

Gene and tissue	Disease dataset	OR (95% CI)	P value	Adjusted P	Statistical support
TSHR Subcutaneous adipose	**FinnGen GD**	0.36 (0.31–0.43)	**1.28×10^{-34}**	**1.28×10^{-33}**	Corrected support*
	BBJ GD	0.46 (0.38–0.55)	**1.92×10^{-15}**	**7.67×10^{-14}**	Corrected support*
	UKB HT	0.31 (0.27–0.37)	**2.86×10^{-42}**	**1.14×10^{-40}**	Corrected support*
	FinnGen GO	0.45 (0.32–0.63)	**5.76×10^{-6}**	**2.30×10^{-4}**	Corrected support*
	FinnGen GO strict	0.51 (0.35–0.74)	**3.67×10^{-4}**	**0.0147**	Corrected support*
TSHR Visceral adipose	**FinnGen GD**	0.50 (0.43–0.58)	**4.09×10^{-19}**	**4.09×10^{-18}**	Corrected support*
	BBJ GD	0.56 (0.46–0.67)	**2.64×10^{-10}**	**1.06×10^{-8}**	Corrected support*
	UKB HT	0.39 (0.34–0.45)	**9.19×10^{-38}**	**3.68×10^{-36}**	Corrected support*
	FinnGen GO	0.69 (0.49–0.98)	0.0368	1.000	Nominal only
	FinnGen GO strict	0.78 (0.54–1.13)	0.186	1.000	Not significant
IGF1R Thyroid	**FinnGen GD**	0.76 (0.56–1.02)	0.0692	0.692	Not significant
	BBJ GD	1.23 (0.90–1.68)	0.200	1.000	Not significant
	UKB HT	1.33 (0.99–1.80)	0.0594	1.000	Not significant
	FinnGen GO	1.02 (0.54–1.94)	0.945	1.000	Not significant
	FinnGen GO strict	1.23 (0.62–2.44)	0.549	1.000	Not significant
IGF1R Cultured fibroblasts	**FinnGen GD**	1.11 (0.89–1.38)	0.359	1.000	Not significant
	BBJ GD	0.91 (0.70–1.20)	0.511	1.000	Not significant
	UKB HT	1.03 (0.82–1.29)	0.815	1.000	Not significant
	FinnGen GO	0.78 (0.49–1.25)	0.301	1.000	Not significant
	FinnGen GO strict	0.78 (0.47–1.30)	0.341	1.000	Not significant

* Bold P values indicate family-corrected support. Primary FinnGen GD: raw $P < 0.005$, adjusted $P = \min(1, 10P)$. Secondary datasets: raw $P < 0.00125$, adjusted $P = \min(1, 40P)$. Nominal only means $P < 0.05$ without passing the relevant corrected threshold. The correction denominators retain unestimable comparisons.

ORs are per unit of normalized, inverse-normal-transformed expression. Effect magnitudes are not directly comparable to reconstructed eQTLGen units. GD, Graves disease; HT, hyperthyroidism; GO, Graves ophthalmopathy. FinnGen endpoints overlap and use population controls. No tissue-specific colocalization was performed. Non-significance does not establish absence of an effect.

TSHR and IGF1R tissue extension

A Instrument availability

GTEx v8 tissue	*TSHR*	*IGF1R*
Thyroid	X	**1**
Cultured fibroblasts	−	**1**
Subcutaneous adipose	**1**	X
Visceral adipose	**1**	X
Whole blood	X	X

1 One qualifying instrument

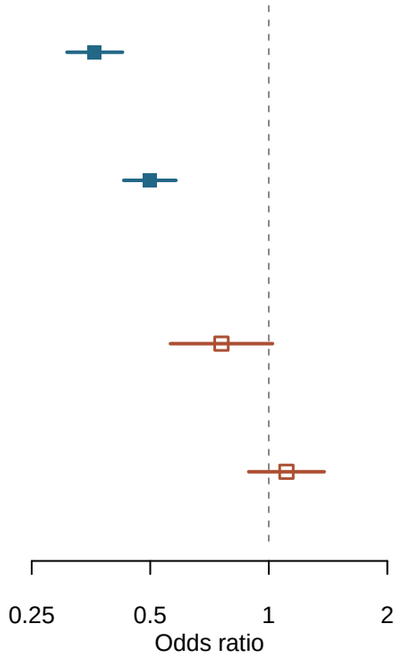
x No eQTL meeting P < 5e−8

− Gene absent from the tested−gene file

B Primary comparison within the tissue extension

Disease dataset: FinnGen Graves disease

Expression instrument	OR (95% CI)	P value	Adjusted P
TSHR Subcutaneous adipose	0.36 (0.31−0.43)	**1.28e−34**	**1.28e−33***
TSHR Visceral adipose	0.50 (0.43−0.58)	**4.09e−19**	**4.09e−18***
IGF1R Thyroid	0.76 (0.56−1.02)	0.0692	0.692
IGF1R Cultured fibroblasts	1.11 (0.89−1.38)	0.359	1.000



* Corrected support: raw P < 0.005; adjusted P = min(1, 10 x P). Filled squares mark corrected support.

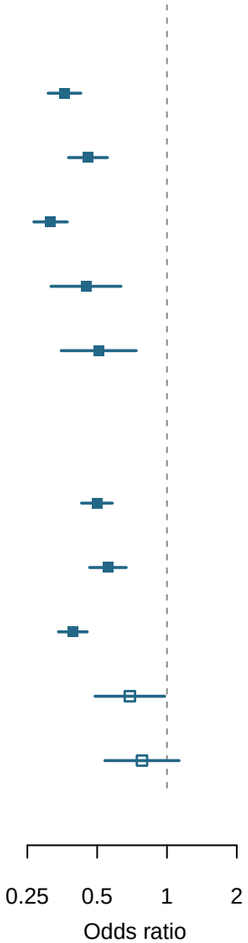
Ten primary combinations were planned; six were unestimable. All 40 secondary comparisons are reported separately.

GTEx blood differs from eQTLGen. ORs are per normalized expression unit. Non−orbital tissues; no tissue−specific colocalization.

Tissue−specific expression associations across five disease datasets

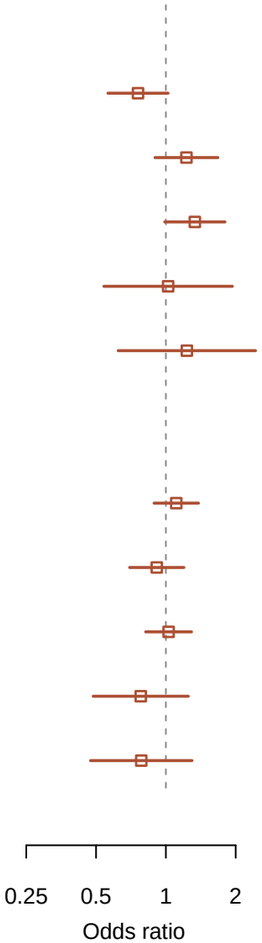
A TSHR

Disease dataset	OR (95% CI)	P value
Subcutaneous adipose		
FinnGen GD	0.36 (0.31−0.43)	**1.28e−34***
BBJ GD	0.46 (0.38−0.55)	**1.92e−15***
UKB hyperthyroidism	0.31 (0.27−0.37)	**2.86e−42***
FinnGen ophthalmopathy	0.45 (0.32−0.63)	**5.76e−06***
FinnGen strict ophthalmopathy	0.51 (0.35−0.74)	**3.67e−04***
Visceral adipose		
FinnGen GD	0.50 (0.43−0.58)	**4.09e−19***
BBJ GD	0.56 (0.46−0.67)	**2.64e−10***
UKB hyperthyroidism	0.39 (0.34−0.45)	**9.19e−38***
FinnGen ophthalmopathy	0.69 (0.49−0.98)	0.0368
FinnGen strict ophthalmopathy	0.78 (0.54−1.13)	0.1863



B IGF1R

Disease dataset	OR (95% CI)	P value
Thyroid		
FinnGen GD	0.76 (0.56−1.02)	0.0692
BBJ GD	1.23 (0.90−1.68)	0.1996
UKB hyperthyroidism	1.33 (0.99−1.80)	0.0594
FinnGen ophthalmopathy	1.02 (0.54−1.94)	0.9449
FinnGen strict ophthalmopathy	1.23 (0.62−2.44)	0.5493
Cultured fibroblasts		
FinnGen GD	1.11 (0.89−1.38)	0.3591
BBJ GD	0.91 (0.70−1.20)	0.5111
UKB hyperthyroidism	1.03 (0.82−1.29)	0.8147
FinnGen ophthalmopathy	0.78 (0.49−1.25)	0.3005
FinnGen strict ophthalmopathy	0.78 (0.47−1.30)	0.3413



Each estimate uses one cis−eQTL. ORs are per unit of normalized, inverse−normal−transformed expression.

* Family−corrected: FinnGen GD P < 0.005; other datasets P < 0.00125. Unmarked estimates do not pass these thresholds.

FinnGen endpoints overlap. Adipose and cultured fibroblasts are not orbital tissue. Thirty unestimable planned comparisons are tabulated separately.